Applications Of Forces

Solutions

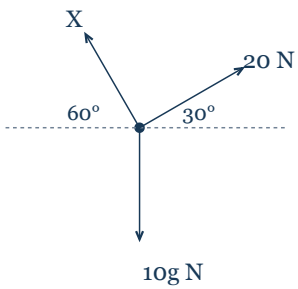
Overview

Solution 2

Combine friction, resolving forces, $F = ma$, inclined planes and connected particles. These preview examples are developed in the chapter.

1. Equilibrium

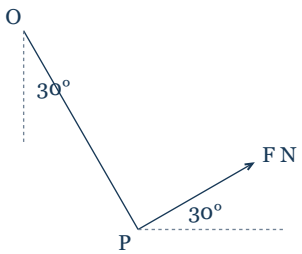
Resolve forces to determine unknown magnitudes.



The supplied numerical sketch is inconsistent with equilibrium for $g = 9.8$; its arrows illustrate resolving

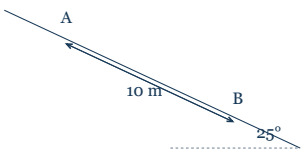
2. Weight and tension

A 2 kg particle on a light string is held at 30° to the downward vertical by a force F, 30° above horizontal. Find F and the tension. (8 marks)



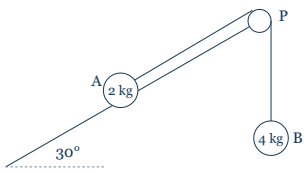
3. Inclined motion

A 0.6 kg particle slides 10 m down a rough 25° plane in 3.5 s, starting at 2 m s^{-1}. Find final speed, acceleration and friction coefficient. (3, 2, 5



4. Connected particles

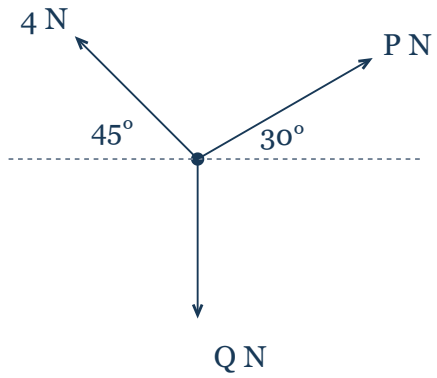
A 2 kg particle on a rough 30° slope is connected by a taut light inextensible string over a smooth fixed pulley to a hanging 4 kg particle. With $\mu = 1/\sqrt{3}$, release from rest makes A move uphill. Find tension immediately after



Finding Unknown Forces by Resolving

Solution 3

The particle is in equilibrium under the forces shown. Resolve horizontally and vertically to find the magnitudes P and Q.



Working

The resultant horizontal force is zero:

$$P\cos 30^\circ - 4\cos 45^\circ = 0.$$

Hence

$$P\cos 30^\circ = 4\cos 45^\circ.$$

Vertically:

$$P\sin 30^\circ + 4\sin 45^\circ = Q.$$

$$P = \frac{4\cos 45^\circ}{\cos 30^\circ} = 3.27\,\text{N} \quad (3\text{ s.f.}).$$

$$Q = \frac{4\cos 45^\circ}{\cos 30^\circ}\sin 30^\circ + 4\sin 45^\circ$$

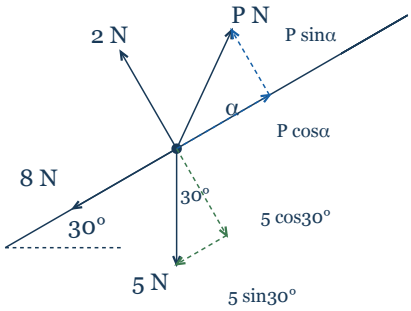
$$Q = 4.46\,\text{N} \quad (3\text{ s.f.}).$$

Resolve separately in the two directions, as before.

Unknown Forces on Inclined Planes

Solution 4

The particle is in equilibrium on the inclined plane under the forces shown. Find P and α.



Use dotted, labelled arrows for resolved components. Blue arrows resolve P; green arrows resolve the weight.

Resolve the applied force:

P parallel Along the plane: $P\cos\alpha$.

P normal Normal to the plane: $P\sin\alpha$.

Resolve the weight:

Weight normal Into the plane: $5\cos 30^\circ$.

Weight parallel Down the plane: $5\sin 30^\circ$.

Working

Along the plane:

$$P\cos\alpha = 8 + 5\sin 30^\circ.$$

Normal to it:

$$P\sin\alpha + 2 = 5\cos 30^\circ.$$

Divide the sine and cosine equations to eliminate P, as in the $R\sin(x+\alpha)$ method:

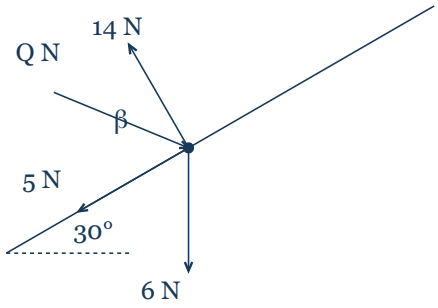
$$\tan\alpha = \frac{5\cos 30^\circ - 2}{8 + 5\sin 30^\circ} = 0.222\ldots$$

$$\alpha = 12.5^\circ \quad (3\text{ s.f.}).$$

$$P\sin\alpha = 5\cos 30^\circ - 2 \quad \Rightarrow \quad P = 10.8\,\text{N} \quad (3\text{ s.f.}).$$

Test Your Understanding — Solution 5

The particle is in equilibrium under the forces shown.
Find Q and β.



Resolve

Along the plane:

$Q\cos\beta = 5 + 6\sin 30°.$

Normal to it:

$Q\sin\beta + 6\cos 30° = 14.$

Solve

$Q\sin\beta = 14 - 6\cos 30°, \quad Q\cos\beta = 5 + 6\sin 30°.$

$$\tan\beta = \frac{14 - 6\cos 30°}{5 + 6\sin 30°}$$

$\beta = 47.7°$ (3 s.f.).

$Q\cos\beta = 5 + 6\sin 30° \quad \Rightarrow \quad Q = 11.9\,\text{N}$ (3 s.f.).

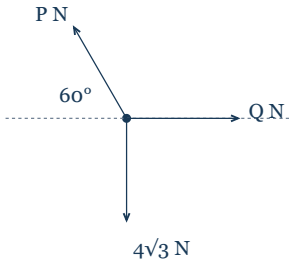
Exercise 7A — Y2 Mechanics Pages 130–132 — Solution 6

2. For each particle in static equilibrium:

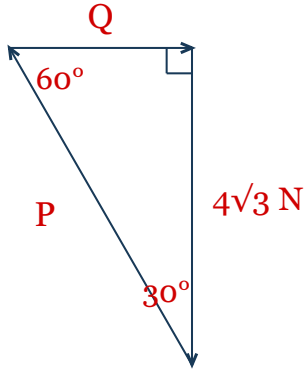
i. Draw a triangle of forces.

ii. Use trigonometry to find unknown forces P, Q and unknown angle θ.

a.



a.i



a.ii

Horizontally, $Q - P\cos 60° = 0.$ (1)

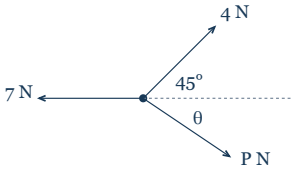
Vertically, $P\sin 60° - 4\sqrt{3} = 0.$ (2)

$$P = \frac{4\sqrt{3}}{\sin 60°} = 4\sqrt{3}\frac{2}{\sqrt{3}} = 8\,\text{N}.$$

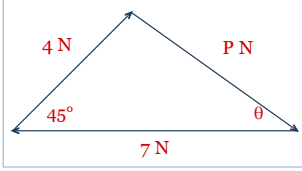
Substitute into (1):

$Q = 8\cos 60° = 8 \times \frac{1}{2} = 4\,\text{N}.$

b.



b.i



b.ii

Horizontally:

$4\cos 45° + P\cos\theta - 7 = 0$

$P\cos\theta = 7 - 4\cos 45°.$ (1)

Vertically:

$4\sin 45° - P\sin\theta = 0$

$P\sin\theta = 4\sin 45°.$ (2)

Divide (2) by (1):

$$\tan\theta = \frac{4\sin 45°}{7 - 4\cos 45°} = \frac{2.828\ldots}{4.172\ldots} = 0.678\ldots$$

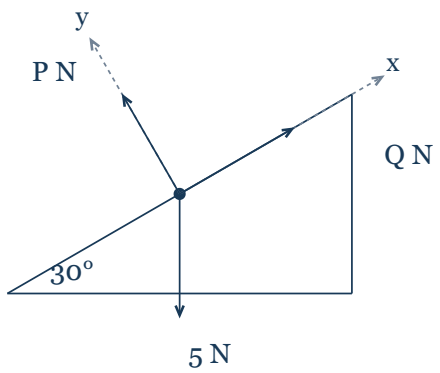
$\theta = 34.1°$ (3 s.f.).

Substitute into (2), using full precision:

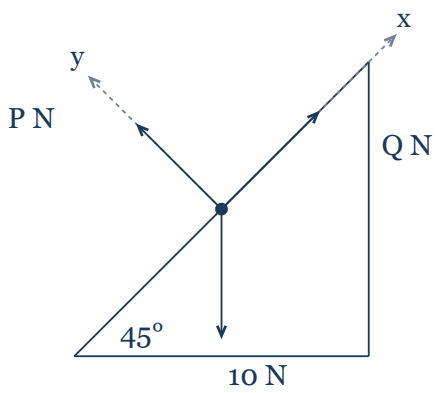
$$P = \frac{2.828\ldots}{\sin\theta} = 5.04\,\text{N} \quad (3\text{ s.f.}).$$

3. Each particle rests in equilibrium on the plane under the forces shown. Find P and Q.

a.



b.



a

Normal to the plane:

$P = 5\cos 30^\circ = 4.33\,\text{N}.$

Along the plane:

$Q = 5\sin 30^\circ = 2.5\,\text{N}.$

b

Along the plane:

$Q - 10\sin 45^\circ = 0. \qquad (1)$

Normal to it:

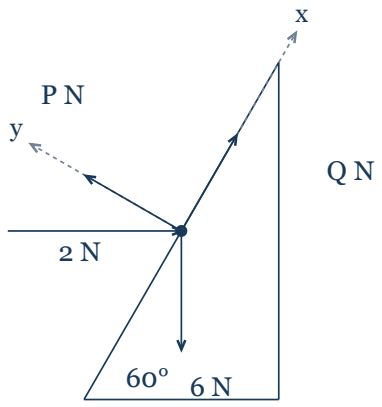
$P - 10\cos 45^\circ = 0. \qquad (2)$

$P = 10\cos 45^\circ = 5\sqrt{2} = 7.07\,\text{N}.$

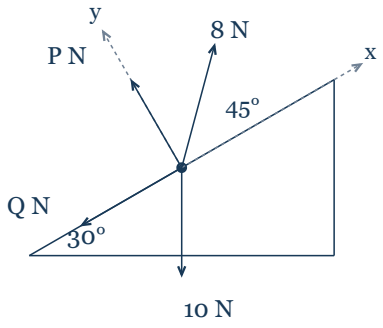
$Q = 10\sin 45^\circ = 5\sqrt{2} = 7.07\,\text{N}.$

Both answers to 3 s.f.

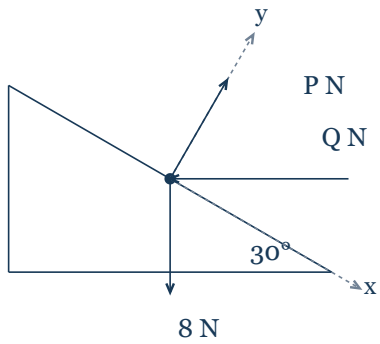
3c.



3d.



3e.



c

Along the plane:

$Q + 2\cos 60^\circ - 6\sin 60^\circ = 0.$

Normal to it:

$P - 2\sin 60^\circ - 6\cos 60^\circ = 0.$

$P = 2\sin 60^\circ + 6\cos 60^\circ$

$P = 4.73\,\text{N} \quad (3\text{ s.f.}).$

$Q = 6\sin 60^\circ - 2\cos 60^\circ$

$Q = 4.20\,\text{N} \quad (3\text{ s.f.}).$

d

Along the plane:

$8\cos 45^\circ - 10\sin 30^\circ - Q = 0.$

Normal to it:

$P + 8\sin 45^\circ - 10\cos 30^\circ = 0.$

$P = 10\cos 30^\circ - 8\sin 45^\circ$

$= 5\sqrt{3} - 4\sqrt{2} = 3.00\,\text{N}.$

$Q = 8\cos 45^\circ - 10\sin 30^\circ$

$= 4\sqrt{2} - 5 = 0.657\,\text{N}.$

Both to 3 s.f.

e

Down the plane:

$8\sin 30^\circ - Q\cos 30^\circ = 0.$

Normal to it:

$P - Q\sin 30^\circ - 8\cos 30^\circ = 0.$

$Q = \dfrac{8\sin 30^\circ}{\cos 30^\circ} = 8\tan 30^\circ$

$= \dfrac{8\sqrt{3}}{3} = 4.62\,\text{N}.$

$P = Q\sin 30^\circ + 8\cos 30^\circ$

$= \dfrac{8\sqrt{3}}{3}\dfrac{1}{2} + 8\dfrac{\sqrt{3}}{2}$

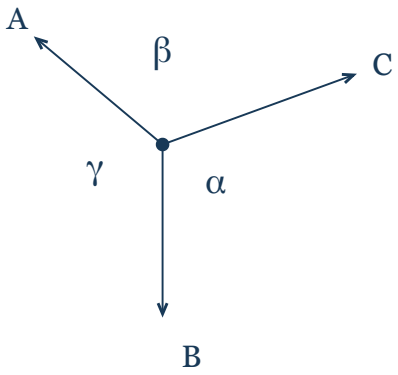
$= \dfrac{4\sqrt{3}}{3} + 4\sqrt{3} = \dfrac{16\sqrt{3}}{3}$

$P = 9.24\,\text{N} \quad (3\text{ s.f.}).$

Challenge

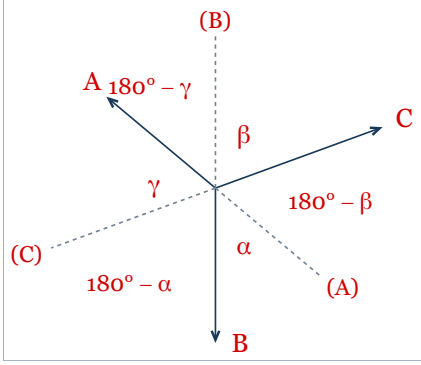
Three coplanar forces of magnitudes A, B, C act on a particle in equilibrium, as shown. Prove

$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}.$$



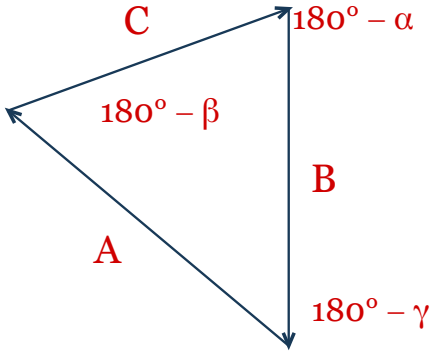
Proof

Extend each line of action backwards through the particle to identify supplementary angles between the lines.



Proof

Equilibrium means the force vectors form a closed triangle.



Proof

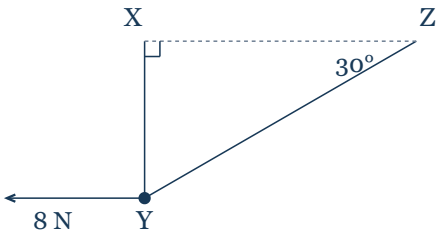
By the sine rule:

$$\frac{A}{\sin(180^\circ - \alpha)} = \frac{B}{\sin(180^\circ - \beta)} = \frac{C}{\sin(180^\circ - \gamma)}.$$

For any angle θ, $\sin(180^\circ - \theta) = \sin\theta$. Hence

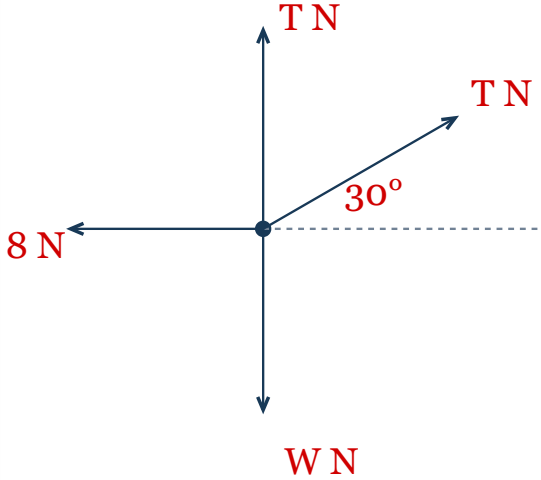
$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}.$$

Apply resolving to tension, weight and pulleys. A smooth bead Y is threaded on a light inextensible string whose ends are fixed at X, Z, at the same horizontal level. A horizontal 8 N force parallel to ZX holds the bead in equilibrium, with Y vertically below X and $\angle XZY = 30^\circ$. Find the tension and the bead's weight.



Draw the force diagram.

Diagram



The smooth bead allows the tension to be the same in both parts of the light string.

Tension

Horizontally:

$T\cos 30^\circ = 8 \quad \Rightarrow \quad T = 9.24\,\text{N} \quad (3\text{ s.f.}).$

Weight

Vertically:

$W = T + T\sin 30^\circ = 13.9\,\text{N} \quad (3\text{ s.f.}).$

Use the unrounded value of T.

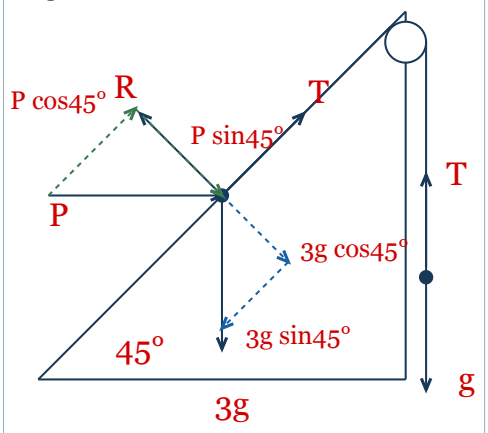
Further Example

Solution 11

A 3 kg mass rests on a smooth plane at 45° to horizontal. A cable runs up the line of greatest slope over a smooth pulley and supports a freely hanging 1 kg mass. Model the masses as particles and the cable as a light inextensible string. A horizontal force P acts towards the plane on the 3 kg mass; the system is in equilibrium. Find: (a) P; (b) the normal reaction; (c) how the smooth-pulley assumption is used.

Draw the force diagram.

Diagram



a.i

For the 1 kg mass, vertically:

$T = g.$

a.ii

For the 3 kg mass, up the plane:

$T + P\cos 45^\circ = 3g\sin 45^\circ.$

$$P = \frac{3g\sin 45^\circ - g}{\cos 45^\circ}$$

$= 3g - g\sqrt{2} = 16\,\text{N} \quad (2\text{ s.f.}).$

b

Normal to the plane:

$R = 3g\cos 45^\circ + P\sin 45^\circ$

$= 6g\frac{\sqrt{2}}{2} - g = 32\,\text{N} \quad (2\text{ s.f.}).$

c

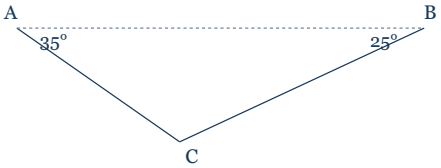
The pulley is smooth, so the tension is the same on both sides of it.

Test Your Understanding

Solution 12

Edexcel M1 (Old), May 2013(R), Q2

A particle of weight 8 N is attached at C to two light inextensible strings AC, BC. Their other ends are fixed to a horizontal ceiling at A, B. The particle rests in equilibrium; the strings lie in a vertical plane, at 35° and 25° to horizontal respectively. Find (i) the tension in AC; (ii) the tension in BC.



The particle cannot slide along the string: these are two separate strings with separate tensions. Introduce T_A, T_B.

Working

Resolve horizontally:

$T_A\cos 35^\circ = T_B\cos 25^\circ.$

Resolve vertically:

$T_A\sin 35^\circ + T_B\sin 25^\circ = 8.$

Eliminate one unknown:

$$T_B\frac{\cos 25^\circ}{\cos 35^\circ}\sin 35^\circ + T_B\sin 25^\circ = 8,$$

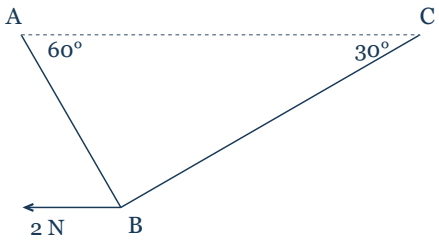
or

$$T_A\sin 35^\circ + T_A\frac{\cos 35^\circ}{\cos 25^\circ}\sin 25^\circ = 8.$$

$T_A = 8.37\,\text{N}, \qquad T_B = 7.57\,\text{N} \quad (3\text{ s.f.}).$

Exercise 7B — Y2 Mechanics Pages 134–137 Solution 13

5. A smooth bead B is threaded on a light inextensible string fixed at A, C, at the same horizontal level. A horizontal 2 N force parallel to CA holds it in equilibrium. The string sections make angles $60°, 30°$ with horizontal. Find (a) the tension; (b) the bead's mass; (c) state how the smooth-bead assumption is used.



a

Let the tension be T and mass be m. Resolve horizontally:

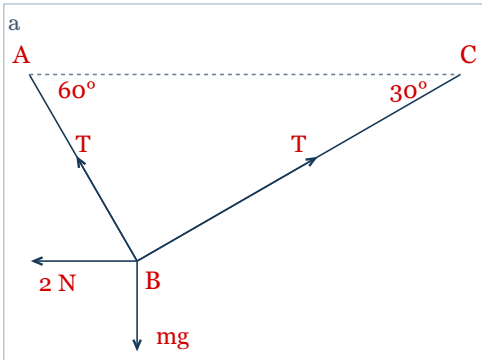
$$T\cos 30° - T\cos 60° - 2 = 0$$

$$T(\cos 30° - \cos 60°) = 2$$

$$T = \frac{2}{\cos 30° - \cos 60°}$$

$$= 5.46\,\text{N} \quad (3\text{ s.f.}).$$

a



b

Vertically:

$$T\sin 60° + T\sin 30° - mg = 0$$

$$mg = T(\sin 60° + \sin 30°)$$

$$m = \frac{2}{g}(\sqrt{3}+1)\left(\frac{\sqrt{3}}{2} + \frac{1}{2}\right)$$

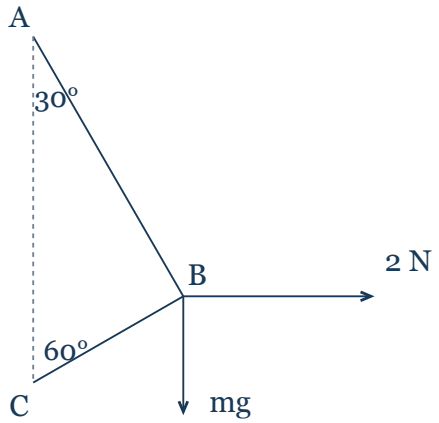
$$= \frac{4+2\sqrt{3}}{g} = 0.76\,\text{kg} \quad (2\text{ s.f.}).$$

c

A smooth bead has no friction with the string, so the tension is the same in both sections.

Exercise 7B — Y2 Mechanics Pages 134–137 Solution 14

6. A smooth bead B is threaded on a light inextensible string fixed at A, C, where A is vertically above C. A horizontal 2 N force holds the bead in equilibrium. Sections AB, BC make angles $30°, 60°$ with vertical respectively. Find (a) the tension; (b) the bead's mass, to the nearest gram.



a

Let tension be T and mass be m. Horizontally:

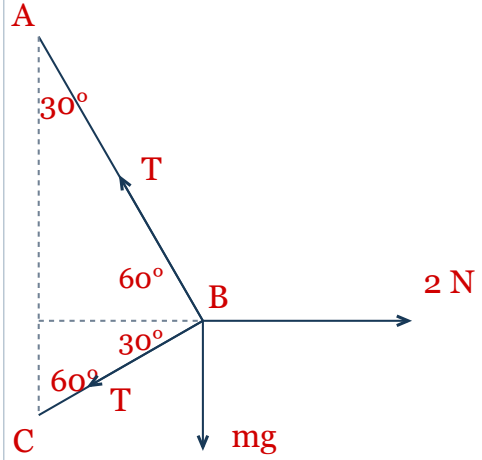
$$2 - T\cos 60° - T\cos 30° = 0$$

$$T(\cos 60° + \cos 30°) = 2$$

$$T = \frac{2}{\cos 60° + \cos 30°}$$

$$= 1.46\,\text{N} \quad (3\text{ s.f.}).$$

a



b

Vertically:

$$T\sin 60° - T\sin 30° - mg = 0.$$

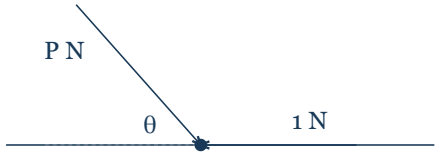
$$m = \frac{4-2\sqrt{3}}{g}$$

$$= 0.05468\ldots\,\text{kg}$$

$$= 55\,\text{g} \quad \text{to the nearest gram.}$$

Exercise 7B — Y2 Mechanics Pages 134–137 Solution 15

7. A particle of weight 2 N rests on a smooth horizontal surface in equilibrium under the two external forces shown. One is a horizontal 1 N force; the other has magnitude P, at angle θ to horizontal, where $\tan\theta = 12/5$. Find (a) P; (b) the normal reaction.



a

$$\tan\theta = \frac{12}{5}$$

$$\Rightarrow \quad \sin\theta = \frac{12}{13}, \quad \cos\theta = \frac{5}{13}.$$

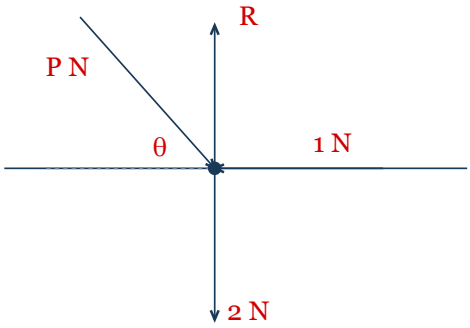
Let the normal reaction be R.

Horizontally:

$$P\cos\theta - 1 = 0$$

$$P = \frac{1}{\cos\theta} = \frac{13}{5} = 2.6\,\text{N}.$$

a



b

Vertically:

$$R - P\sin\theta - 2 = 0$$

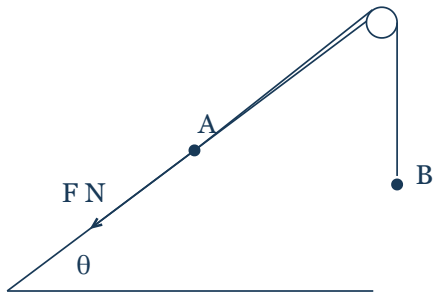
$$R = P\sin\theta + 2$$

$$= 2.6 \times \frac{12}{13} + 2$$

$$= 2.4 + 2 = 4.4\,\text{N}.$$

Exercise 7B — Y2 Mechanics Pages 134–137 Solution 16

11. Particle A, mass 2 kg, rests in equilibrium on a smooth plane at angle θ to horizontal, where $\tan\theta = 3/4$. A light inextensible string runs from A over a smooth pulley to hanging particle B, mass 5 kg. A force F acts on A down the line of greatest slope. Find (a) the normal reaction; (b) F; (c) state how the smooth-pulley assumption is used.



a

Let the normal reaction be R, tension T. For the 5 kg mass:

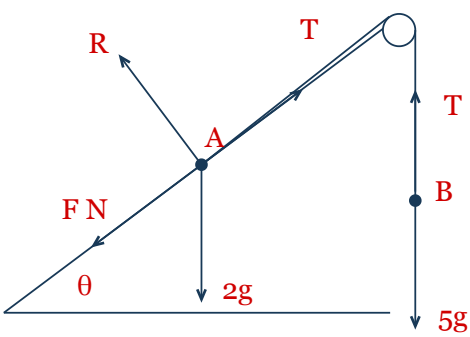
$$T - 5g = 0 \quad \Rightarrow \quad T = 5g.$$

For the 2 kg mass, normal to plane:

$$R - 2g\cos\theta = 0$$

$$R = 2g \times \frac{4}{5} = 16\,\text{N} \quad (2\text{ s.f.}).$$

a



b

Up the plane:

$$T - F - 2g\sin\theta = 0$$

$$F = T - 2g\sin\theta = 5g - 2g \times \frac{3}{5}$$

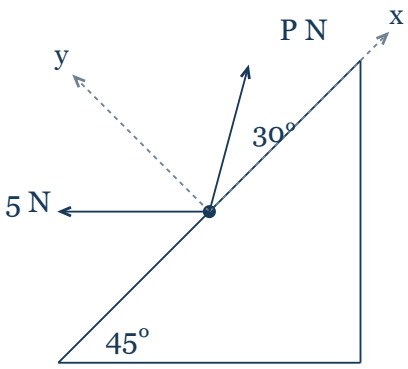
$$= \frac{19g}{5} = 37\,\text{N} \quad (2\text{ s.f.}).$$

c

A smooth pulley has no friction with the string, so the tension is the same on each side.

Exercise 7B — Y2 Mechanics Pages 134–137 Solution 17

12. A particle of weight 20 N rests in equilibrium on a smooth 45° inclined plane. A horizontal 5 N force acts to the left; force P acts at 30° above the plane. Find the normal reaction.



Working

Let the normal reaction be R. Resolve along the plane first: P is the only unknown in that direction.

$$P\cos 30^\circ - 5\cos 45^\circ - 20\sin 45^\circ = 0$$

$$P = \frac{5\cos 45^\circ + 20\sin 45^\circ}{\cos 30^\circ}$$

$$= 20.4\,\text{N} \quad (3\text{ s.f.}).$$

Normal to the plane:

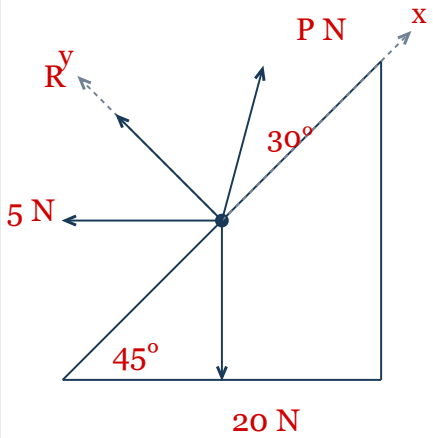
$$R + P\sin 30^\circ + 5\sin 45^\circ - 20\cos 45^\circ = 0$$

$$R = 20\cos 45^\circ - 5\sin 45^\circ - P\sin 30^\circ$$

$$= 0.400\,\text{N} \quad (3\text{ s.f.}).$$

Use unrounded P.

Working

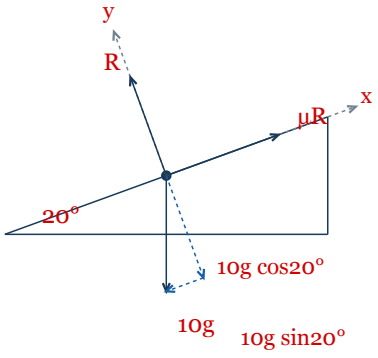


Friction Solution 18

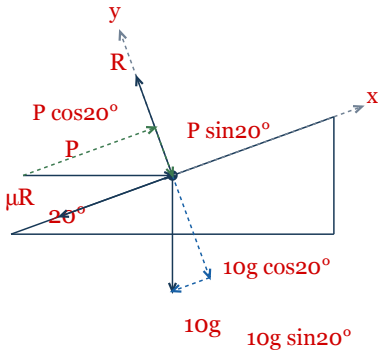
Friction satisfies $F \le \mu R$, with $F = \mu R$ in limiting equilibrium or sliding under this model. At rest, friction opposes the motion that would occur without it.

A 10 kg box rests in limiting equilibrium on a rough plane at 20° above horizontal. (a) Find μ. A horizontal force P is then applied towards the plane. (b) Find the maximum P for equilibrium.

a diagram



b diagram



a working

Normal to plane:

$$R = 10g\cos 20^\circ.$$

Up the plane:

$$\mu R = 10g\sin 20^\circ.$$

$$\mu = \frac{10g\sin 20^\circ}{10g\cos 20^\circ} = \tan 20^\circ$$

$$= 0.36 \quad (2\text{ s.f.}).$$

b working

At maximum P, impending motion is uphill, so friction acts downhill.

$$R = 10g\cos 20^\circ + P\sin 20^\circ$$

$$P\cos 20^\circ = 10g\sin 20^\circ + \mu R$$

$$P\cos 20^\circ = 10g\sin 20^\circ + \tan 20^\circ(10g\cos 20^\circ + P\sin 20^\circ).$$

Collect terms:

$$P = \frac{20g\sin 20^\circ}{\cos 20^\circ - \tan 20^\circ\sin 20^\circ}$$

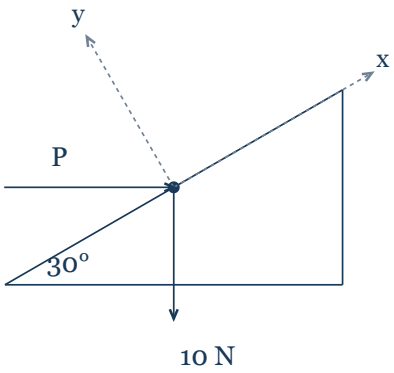
$$= 82\,\text{N} \quad (2\text{ s.f.}).$$

Test Your Understanding

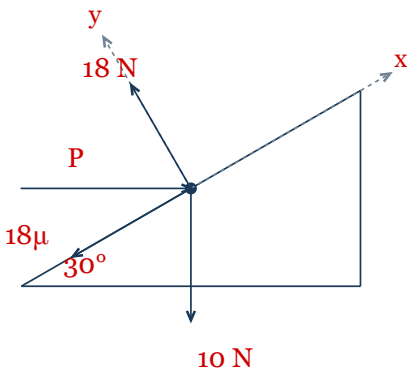
Solution 19

Edexcel M1 (Old), January 2006, Q5

A parcel of weight 10 N lies on a rough 30° inclined plane. Horizontal force P acts towards the plane. It is in equilibrium on the point of slipping uphill; the normal reaction is 18 N, coefficient of friction μ. Find (a) P (4 marks); (b) μ (5 marks). The horizontal force is removed. (c) Determine whether the parcel moves (5 marks).



a



b

Parallel to plane:

$P\cos 30^\circ = 10\sin 30^\circ + F.$

$F = 18\mu.$

$$\mu = \frac{P\cos 30^\circ - 10\sin 30^\circ}{18}$$

$= 0.621 \quad (3\text{ s.f.}).$

Use unrounded P.

c

Now $R_{\text{new}} = 10\cos 30^\circ$. The weight component downhill is $10\sin 30^\circ = 5$ N.

Maximum available friction:

$F_{\max} = \mu R_{\text{new}} = 5.37$ N.

Since $5.37 > 5$, friction can balance the weight component: the parcel does not slide.

a

Normal to plane:

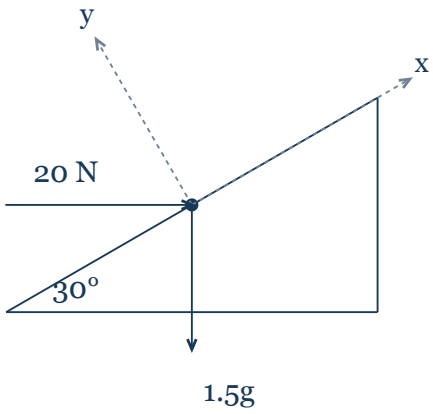
$P\sin 30^\circ + 10\cos 30^\circ = 18$

$P = 18.7$ N $\quad (3\text{ s.f.}).$

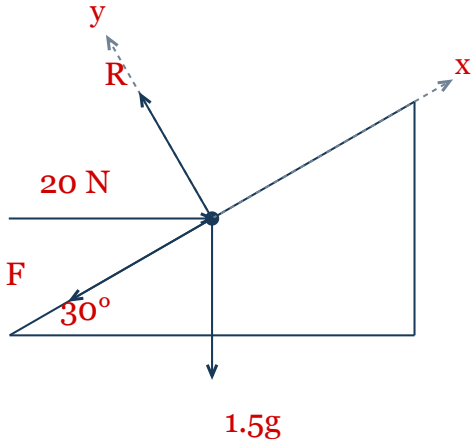
Exercise 7C — Y2 Mechanics Pages 139–142

Solution 20

8. A horizontal 20 N force acts on a 1.5 kg block resting in equilibrium on a rough 30° plane. The force acts in the vertical plane containing the line of greatest slope. Find (a) the normal reaction; (b) the friction magnitude and direction; (c) the minimum coefficient of friction.



Working



Working

Let the normal reaction be R and friction F act downhill.

Normal to the plane:

$R - 20\sin 30^\circ - 1.5g\cos 30^\circ = 0$

$R = 20\sin 30^\circ + 1.5g\cos 30^\circ$

$= 22.7$ N $\quad (3\text{ s.f.}).$

Parallel to it:

$20\cos 30^\circ - F - 1.5g\sin 30^\circ = 0$

$F = 20\cos 30^\circ - 1.5g\sin 30^\circ$

$= 9.97$ N $\quad (3\text{ s.f.}),$ downhill.

Working

The minimum μ occurs when required friction equals μR:

$F = \mu R$

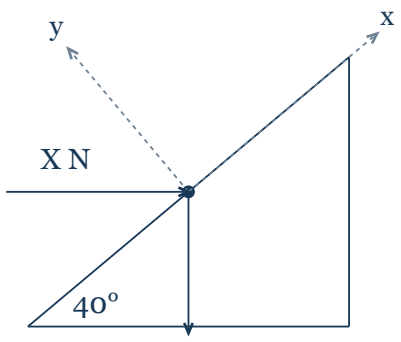
$9.9705\ldots = \mu \times 22.730\ldots$

$$\mu = \frac{9.9705\ldots}{22.730\ldots} = 0.43863\ldots$$

The coefficient must be at least 0.439 (3 s.f.) to prevent sliding.

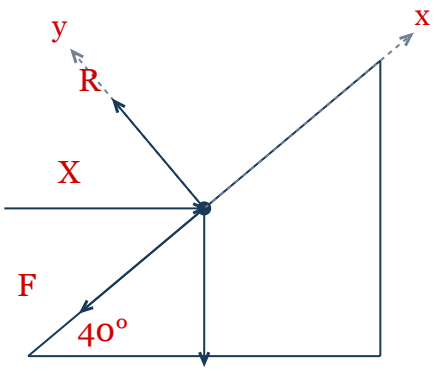
Exercise 7C — Y2 Mechanics Pages 139–142 Solution 21

9. A 3 kg box lies on a rough plane inclined at 40° to horizontal. A horizontal force X holds it in equilibrium, acting in the vertical plane containing the line of greatest slope. The coefficient of friction is 0.3; the box is in limiting equilibrium about to move uphill. Find (a) X; (b) the normal reaction.



3g

a



3g

a

Normal to plane:

$R - X\sin 40^\circ - 3g\cos 40^\circ = 0$

$R = X\sin 40^\circ + 3g\cos 40^\circ.$ (1)

Parallel to plane:

$X\cos 40^\circ - F - 3g\sin 40^\circ = 0$

$F = X\cos 40^\circ - 3g\sin 40^\circ.$ (2)

Limiting friction: $F = \mu R$.

Substitute (1),(2):

$X\cos 40^\circ - 3g\sin 40^\circ$

$= 0.3(X\sin 40^\circ + 3g\cos 40^\circ).$

a

$X\cos 40^\circ - 0.3X\sin 40^\circ$

$= 0.9g\cos 40^\circ + 3g\sin 40^\circ$

$X(\cos 40^\circ - 0.3\sin 40^\circ)$

$= 0.9g\cos 40^\circ + 3g\sin 40^\circ$

$X = \dfrac{0.9g\cos 40^\circ + 3g\sin 40^\circ}{\cos 40^\circ - 0.3\sin 40^\circ}$

$= \dfrac{25.65\ldots}{0.5732\ldots} = 44.8\,\text{N}$ (3 s.f.).

b

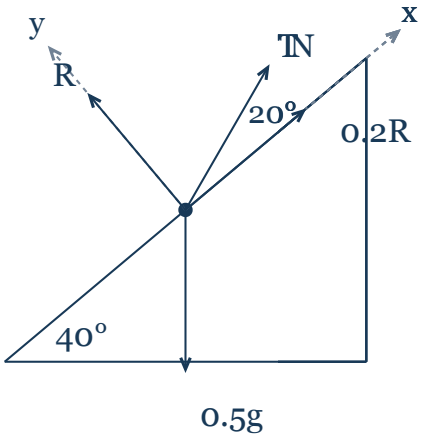
Substitute the unrounded X in (1):

$R = X\sin 40^\circ + 3g\cos 40^\circ$

$= 51.3\,\text{N}$ (3 s.f.).

Exercise 7C — Y2 Mechanics Pages 139–142 Solution 22

11. A 0.5 kg box lies on a 40° inclined plane with coefficient of friction 1/5. A light string keeps it in equilibrium, lying in the vertical plane containing the line of greatest slope, at 20° above the plane. Model the box as a particle. The tension is T. Find its possible range, with limiting equilibrium at the endpoints.



0.5g

Maximum

Normal to the plane:

$R - 0.5g\cos 40^\circ + T\sin 20^\circ = 0$

$R = 0.5g\cos 40^\circ - T\sin 20^\circ.$

At $T_{\max}$, impending motion is uphill and $F = \mu R$ acts downhill:

$T_{\max}\cos 20^\circ - 0.5g\sin 40^\circ$

$= F$

$T_{\max}\cos 20^\circ - 0.5g\sin 40^\circ$

$-\frac{1}{5}(0.5g\cos 40^\circ)$

$+\frac{1}{5}T_{\max}\sin 20^\circ = 0$

$T_{\max} = \dfrac{0.5g\sin 40^\circ + 0.1g\cos 40^\circ}{\cos 20^\circ + 0.2\sin 20^\circ}$

$= 3.8690\ldots\ \text{N}.$

Minimum

At $T_{\min}$, impending motion is downhill and $F = \mu R$ acts uphill:

$T_{\min}\cos 20^\circ - 0.5g\sin 40^\circ$

$= -F$

$T_{\min}\cos 20^\circ - 0.5g\sin 40^\circ$

$+\frac{1}{5}(0.5g\cos 40^\circ)$

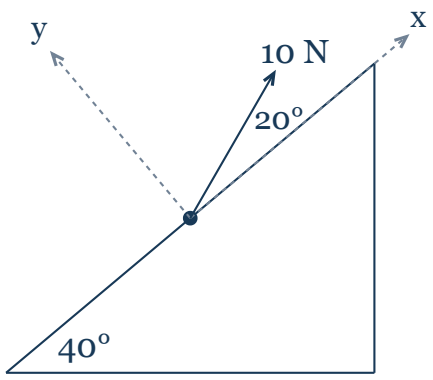
$-\frac{1}{5}T_{\min}\sin 20^\circ = 0$

$T_{\min} = \dfrac{0.5g\sin 40^\circ - 0.1g\cos 40^\circ}{\cos 20^\circ - 0.2\sin 20^\circ}$

$= 2.7533\ldots\ \text{N}.$

Thus $2.75 \le T \le 3.87$ N, with endpoints rounded to 3 s.f.

12. A 1 kg box lies on a plane at 40° to horizontal. A light string, at 20° above the plane in the vertical plane containing its line of greatest slope, holds the box in equilibrium on the point of moving uphill. Model it as a particle. The tension is 10 N. Find the coefficient of friction μ.



Working

Normal to the plane:

$R + 10\sin 20° - g\cos 40° = 0$

$R = g\cos 40° - 10\sin 20° = 4.087\ldots$ N.

Parallel to the plane:

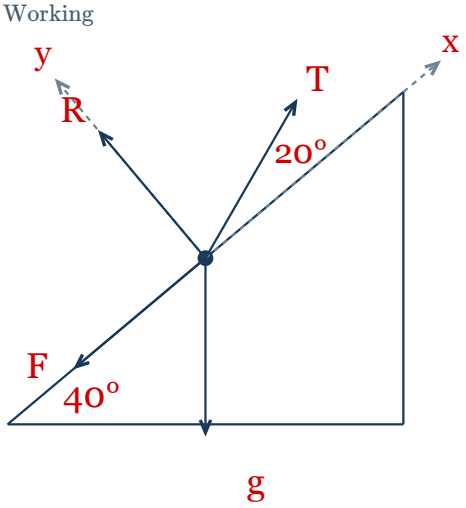
$10\cos 20° - F - g\sin 40° = 0$

$F = 10\cos 20° - g\sin 40° = 3.0976\ldots$ N.

Friction is limiting, $F = \mu R$:

$$\mu = \frac{F}{R} = \frac{3.0976\ldots}{4.087\ldots} = 0.758 \quad (3\text{ s.f.}).$$

Working



13. A 2 kg particle rests in limiting equilibrium on a rough plane at angle θ above horizontal, where $\tan\theta = 3/4$. A horizontal force P, directed into the plane, is then applied. Given that the particle remains in equilibrium, find the maximum possible P.

Working

$\tan\theta = \frac{3}{4} \Rightarrow \sin\theta = \frac{3}{5}, \quad \cos\theta = \frac{4}{5}.$

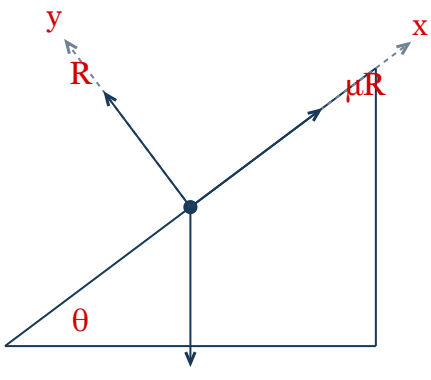
Before P is applied, impending motion is downhill; limiting friction acts uphill.

$R = 2g\cos\theta$

$2g\sin\theta = F_{\max} = \mu R = \mu \times 2g\cos\theta$

$\mu = \tan\theta = \frac{3}{4}.$

Working



Working

At maximum P, impending motion is uphill, so friction acts downhill.

$R' = 2g\cos\theta + P\sin\theta$

$P\cos\theta = F_{\max} + 2g\sin\theta$

$P\cos\theta = \mu R' + 2g\sin\theta$

$P\cos\theta = \mu(2g\cos\theta + P\sin\theta) + 2g\sin\theta$

$P(\cos\theta - \mu\sin\theta) = 2g(\mu\cos\theta + \sin\theta)$

$$P\left(\frac{4}{5} - \frac{3}{4}\times\frac{3}{5}\right) = 2\times 9.8\left(\frac{3}{4}\times\frac{4}{5} + \frac{3}{5}\right)$$

$0.35P = 23.52 \quad \Rightarrow \quad P = 67.2$ N.

The maximum is 67.2 N.

Static Rigid Bodies

Solution 25

For a stationary rigid body:

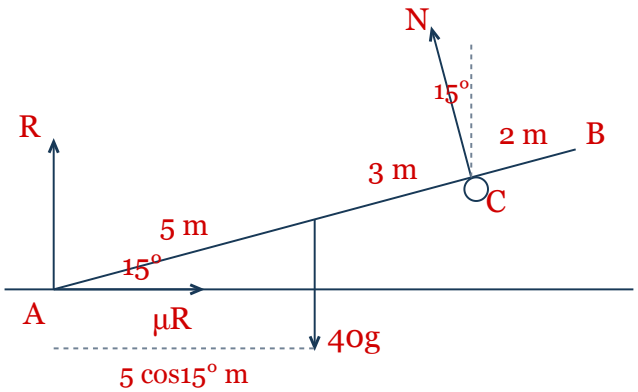
• The resultant force is zero.

• The resultant moment is zero.

These are the moment problems already studied, now including friction.

A uniform rod AB, mass 40 kg, length 10 m, has end A on rough horizontal ground and rests against a smooth peg C, where $AC = 8$ m. It is in limiting equilibrium at 15° to horizontal. Find (a) the reaction at C; (b) the ground's coefficient of friction.

Diagram



a. Take moments about A.

a

$40g \times 5\cos 15° = N \times 8$

$N = 236.65\ldots$ N.

The reaction at C has magnitude 240 N (2 s.f.).

b. Resolve horizontally, then vertically.

b.i $\mu R = N \sin 15° = 61.25$ N

Diagram

The frictional force prevents the rod from sliding along the ground.

b.ii

$R + N\cos 15° = 40g$

$R = 40g - N\cos 15° = 163.41\ldots$ N

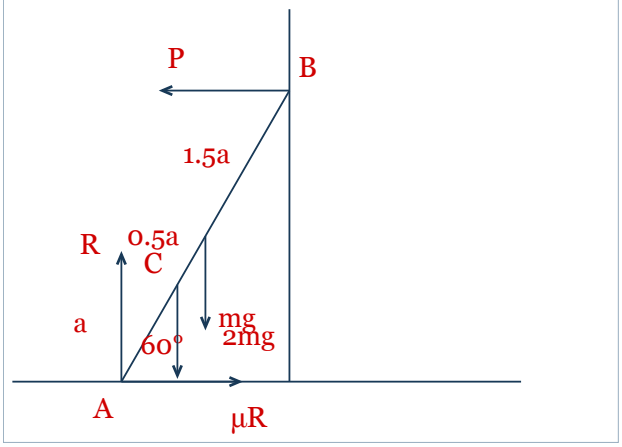
$\mu = \dfrac{61.25}{163.41\ldots} = 0.37$ (2 s.f.).

Further Example

Solution 26

A ladder AB, mass m, length $3a$, rests with A on rough horizontal ground and B against a smooth vertical wall. A load of mass $2m$ is fixed at C, where $AC = a$. Model the ladder as a uniform rod in a vertical plane perpendicular to the wall, and the load as a particle. The ladder rests in limiting equilibrium at 60° to the ground. (a) Find the coefficient of friction with the ground. (b) State how the uniform-ladder assumption is used.

Diagram



Two different reaction forces need different variables.

a.i Horizontally:

$\mu R = P.$

a.ii Vertically:

$R = 2mg + mg = 3mg.$

a.iii

Take moments about B:

$2mg \times 2a\cos 60° + mg \times 1.5a\cos 60°$

$+ \mu R \times 3a\sin 60° = R \times 3a\cos 60°.$

Divide by a:

$2.75mg + \dfrac{3\sqrt{3}}{2}\mu R = 1.5R.$

Substitute $R = 3mg$:

$2.75mg + \dfrac{3\sqrt{3}}{2}\mu(3mg) = 1.5(3mg)$

$2.75 + \dfrac{9\sqrt{3}}{2}\mu = 4.5$

$\mu = 0.225$ (3 s.f.).

b The ladder's weight acts at its midpoint.

Exercise 7D — Y2 Mechanics Pages 144–146 Solution 27

4. A uniform ladder, mass 20 kg, length 8 m, rests against a smooth vertical wall with its lower end on rough horizontal ground. The ground's coefficient of friction is 0.3; the ladder makes angle θ with horizontal, where $\tan\theta = 2$. A 30 kg boy climbs it. Model the ladder as a uniform rod and the boy as a particle. (a) How far can he climb before slipping occurs? (b) Criticise the model for (i) the ladder, (ii) the wall.

a

Suppose the boy reaches B, distance x from the foot A, while the ladder remains in contact with the ground.

$F = N, \qquad R = 50g.$

Moments about A:

$20g \times 4\cos\theta + 30g \times x\cos\theta$

$= N \times 8\sin\theta$

$80g + 30gx = 8N\tan\theta$

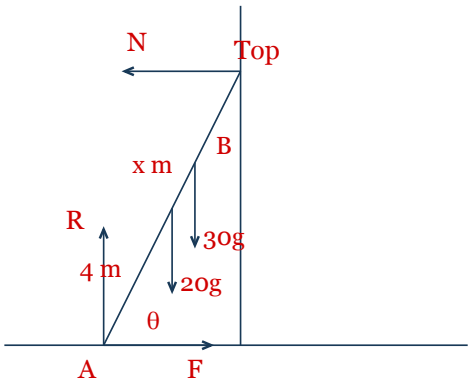
$N = F = \mu R = \dfrac{80g + 30gx}{16}.$

Here $\tan\theta = 2$.

$0.3 \times 50g = \dfrac{80g + 30gx}{16}$

$240 = 80 + 30x \quad \Rightarrow \quad x = 5\tfrac{1}{3}$ m.

a



b.i The ladder may not be uniform.

b.ii There may be friction between the ladder and wall.

Exercise 7D — Y2 Mechanics Pages 144–146 Solution 28

8. A uniform ladder rests in limiting equilibrium with one end on rough horizontal ground and the other against a rough vertical wall. The coefficients of friction are μ_1 at the ground and μ_2 at the wall. It makes angle θ with horizontal. Show

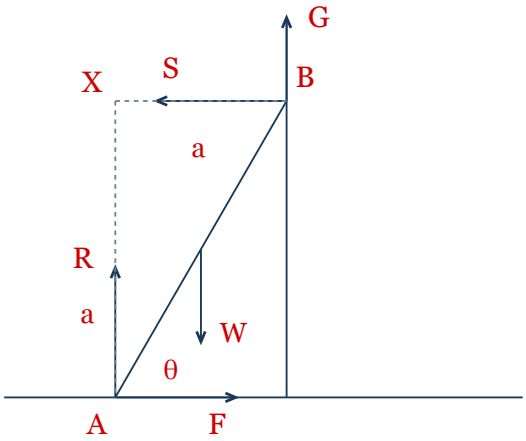
$$\tan\theta = \frac{1 - \mu_1\mu_2}{2\mu_1}.$$

Working

Let R, F be the ground's normal reaction and friction; S, G the wall's reaction and friction; and X the intersection of the lines of action of R, S. Let length be $2a$, weight W. Limiting friction gives

$F = \mu_1 R, \qquad G = \mu_2 S.$

Working



Working

Take moments about X:

$Wa\cos\theta = F \times 2a\sin\theta + G \times 2a\cos\theta$

$W = 2F\tan\theta + 2G. \qquad (1)$

Horizontally, $F = S$; vertically, $W = R + G$.

Substitute for W, F in (1):

$R + G = 2\mu_1 R\tan\theta + 2G$

$R - G = 2\mu_1 R\tan\theta$

$R - \mu_1\mu_2 R = 2\mu_1 R\tan\theta$

$\dfrac{1 - \mu_1\mu_2}{2\mu_1} = \tan\theta.$

Exercise 7D — Y2 Mechanics Pages 144–146 Solution 29

9. A uniform ladder of weight W rests in equilibrium on rough horizontal ground against a smooth vertical wall. Its vertical plane is perpendicular to the wall; the ladder makes 60° with horizontal. The ground's coefficient of friction is μ. (a) Find the wall force in terms of W. (b) Show that $\mu \geq \sqrt{3}/6$.

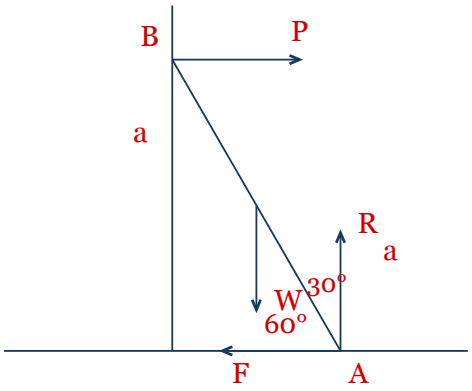
a

Let the ends be A, B, wall reaction P, ground reaction R, ground friction F, and length $2a$. Take moments about A:

$$W \times a\cos 60^\circ = P \times 2a\cos 30^\circ$$

$$P = \frac{W a\cos 60^\circ}{2a\cos 30^\circ} = \frac{W}{2\sqrt{3}}. \qquad (1)$$

a



b

Vertically, $R = W$. (2)

Horizontally, $F = P$. (3)

Equilibrium requires $F \leq \mu R$, hence

$$P \leq \mu R \quad \text{by (3)},$$

$$\frac{W}{2\sqrt{3}} \leq \mu R \quad \text{by (1)},$$

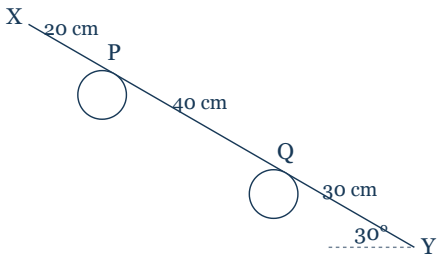
$$\frac{W}{2\sqrt{3}} \leq \mu W \quad \text{by (2)}.$$

Therefore

$$\mu \geq \frac{\sqrt{3}}{6}.$$

Exercise 7D — Y2 Mechanics Pages 144–146 Solution 30

10. A uniform rod XY, weight 20 N, length 90 cm, rests on two parallel pegs with X above Y, in a vertical plane perpendicular to the peg axes. It makes 30° with horizontal and touches the pegs at P, Q, where $XP = 20$ cm, $XQ = 60$ cm. (a) Find the normal force components at P, Q. The friction coefficient at each peg is μ. (b) Given impending slip, find μ.



a

Let T, G be the normal and friction forces at P; S, F those at Q. Moments about P:

$$S \times 40 = 20 \times 25\cos 30^\circ$$

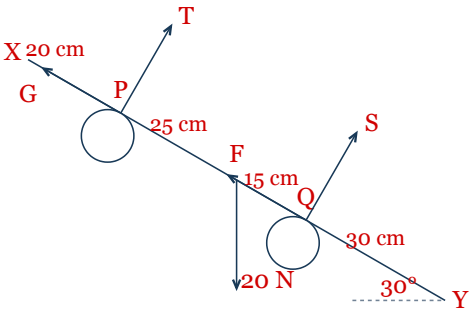
$$S = \frac{20 \times 25 \times \sqrt{3}/2}{40} = \frac{25\sqrt{3}}{4}\ \text{N}.$$

Moments about Q:

$$T \times 40 = 20 \times 15\cos 30^\circ$$

$$T = \frac{20 \times 15 \times \sqrt{3}/2}{40} = \frac{15\sqrt{3}}{4}\ \text{N}.$$

a



b

Along the rod:

$$G + F = 20\cos 60^\circ = 10. \qquad (1)$$

At impending slip, friction is limiting:

$$G = \mu T, \qquad F = \mu S.$$

Using (a):

$$G + F = \mu T + \mu S$$

$$= \mu\frac{40\sqrt{3}}{4} = 10\sqrt{3}\mu. \qquad (2)$$

Equate (1),(2):

$$\mu = \frac{1}{\sqrt{3}}.$$

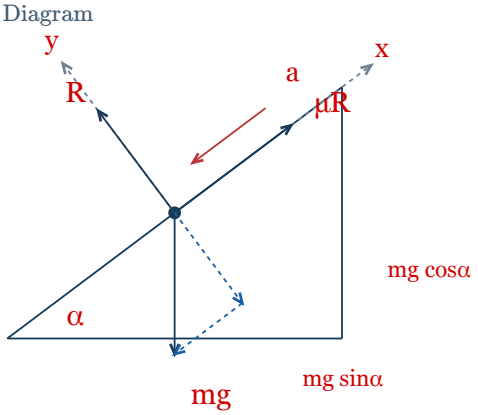
Particles Moving on a Rough Plane

Solution 31

We have considered motion on smooth planes and equilibrium on rough planes. Now consider motion on rough planes.

A particle is held at rest on a rough plane inclined at α to horizontal, where $\tan\alpha = 3/4$. The coefficient of friction is 0.5. Released, it slides downhill. Find (a) its acceleration; (b) the distance travelled in the first 2 s.

Diagram



a

$\tan\alpha = \dfrac{3}{4}$

$\Rightarrow \sin\alpha = \dfrac{3}{5}, \quad \cos\alpha = \dfrac{4}{5}.$

Normal to plane:

$R = mg\cos\alpha = \dfrac{4}{5}mg.$

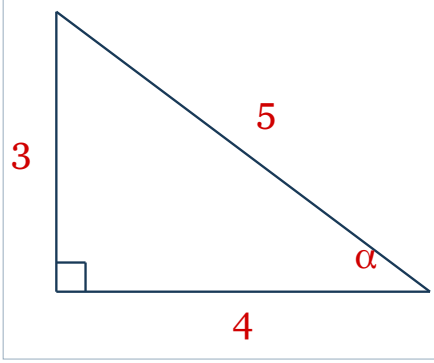
Down the plane:

$mg\sin\alpha - \mu R = ma$

$\dfrac{3}{5}mg - \dfrac{2}{5}mg = ma$

$a = 0.2g = 2.0\,\text{m s}^{-2} \quad (2\text{ s.f.}).$

a



b

$u = 0,\ a = 0.2g,\ t = 2$. Find s:

$s = ut + \frac{1}{2}at^2$

$= \frac{1}{2}(0.2g) \times 2^2$

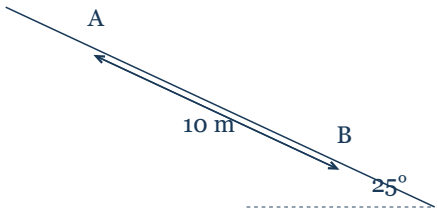
$= 3.9\,\text{m} \quad (2\text{ s.f.}).$

Test Your Understanding

Solution 32

Edexcel M1 (Old), May 2013(R), Q5

Particle P, mass 0.6 kg, slides with constant acceleration down the line of greatest slope of a rough plane at 25° to horizontal. It passes A, B, where $AB = 10$ m, taking 3.5 s from A to B. Its speed at A is $2\,\text{m s}^{-1}$. Find (a) its speed at B; (b) its acceleration; (c) the coefficient of friction.



a

$s = \dfrac{u+v}{2}t$

$10 = \dfrac{2+v}{2} \times 3.5$

$v = \dfrac{20}{3.5} - 2 = \dfrac{26}{7}$

$= 3.71\,\text{m s}^{-1} \quad (3\text{ s.f.}).$

b

$a = \dfrac{v-u}{t} = \dfrac{26/7 - 2}{3.5}$

$= \dfrac{24}{49} = 0.490\,\text{m s}^{-2} \quad (3\text{ s.f.}).$

c

Normal reaction:

$R = 0.6g\cos 25°.$

Parallel to plane:

$0.6g\sin 25° - \mu R = 0.6a$

$\mu = \dfrac{g\sin 25° - a}{g\cos 25°}$

$= 0.411 \quad (3\text{ s.f.}).$

Exercise 7E — Y2 Mechanics Pages 148–150 Solution 33

7. A 10 kg parcel is released from rest on a rough plane inclined at 25° to horizontal. (a) Find the normal reaction. After 2 s, it has moved 4 m downhill. (b) Find the coefficient of friction.

a

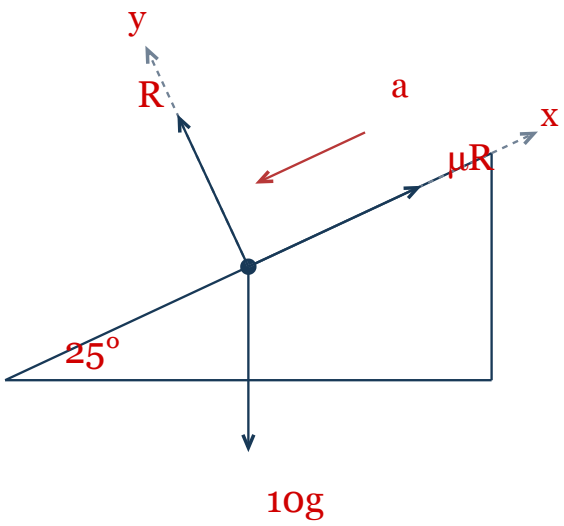
Normal to plane:

$R - 10g\cos 25^\circ = 0$

$R = 98\cos 25^\circ = 88.8\,\text{N}$ (3 s.f.).

The normal reaction is 88.8 N.

a



b

$u = 0, s = 4, t = 2$; find a.

$s = ut + \frac{1}{2}at^2$

$4 = 0 + \frac{1}{2}a \times 2^2 \quad \Rightarrow \quad a = 2\,\text{m s}^{-2}$.

Down the plane, $F_{\text{resultant}} = ma$:

$10g\sin 25^\circ - \mu R = 10 \times 2$

$\mu \times 98\cos 25^\circ = 10g\sin 25^\circ - 20$

$\mu = \dfrac{98\sin 25^\circ - 20}{98\cos 25^\circ} = 0.241$ (3 s.f.).

The coefficient is 0.24 (2 s.f.).

Exercise 7E — Y2 Mechanics Pages 148–150 Solution 34

8. Particle P is projected up a rough plane at angle α to horizontal, where $\tan\alpha = 3/4$. The friction coefficient is $1/3$. From A, initial speed is $20\,\text{m s}^{-1}$; it stops momentarily at B. (a) Show its uphill deceleration is $13g/15$. Find (b) AB, (c) the time from A to B, both to 2 s.f.; (d) its speed on returning to A.

a

Let mass be m. Normal to plane:

$R - mg\cos\alpha = 0$

$\Rightarrow \quad R = \dfrac{4mg}{5}$.

Take uphill as positive:

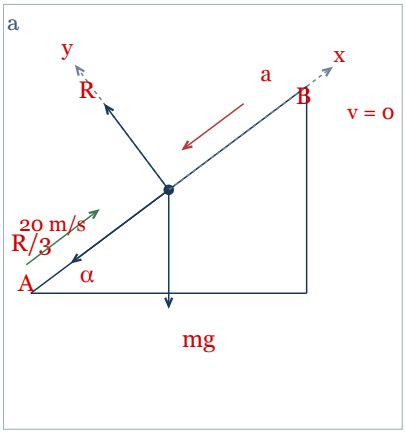
$-mg\sin\alpha - \frac{1}{3}R = ma$

$-\dfrac{3mg}{5} - \frac{1}{3}\dfrac{4mg}{5} = ma$

$a = -\dfrac{13g}{15}$.

The deceleration is $13g/15$.

a



b

$u = 20, v = 0, a = -13g/15$; find s.

$v^2 = u^2 + 2as$

$0 = 20^2 - \dfrac{26g}{15}s$

$s = \dfrac{6000}{26g} = 23.5\ldots\,\text{m}$

$AB = 24\,\text{m}$ (2 s.f.).

c

$v = u + at$

$0 = 20 - \dfrac{13g}{15}t$

$t = \dfrac{300}{13g} = 2.35\ldots\,\text{s}$

$t = 2.4\,\text{s}$ (2 s.f.).

d

On moving downhill from B, friction acts uphill. $R = 4mg/5$ as before. Take downhill positive:

$mg\sin\alpha - \frac{1}{3}R = ma$

$\dfrac{3mg}{5} - \frac{1}{3}\dfrac{4mg}{5} = ma$

$a = \dfrac{g}{3}$.

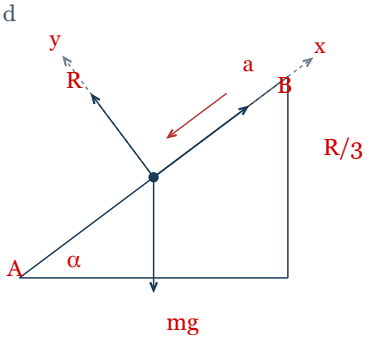
Now $u = 0$, $s = 6000/(26g)$:

$v^2 = u^2 + 2as$

$v^2 = 0 + \dfrac{2g}{3}\dfrac{6000}{26g} = \dfrac{4000}{26}$

$v = 12.4\,\text{m s}^{-1}$ (3 s.f.).

d



Exercise 7E — Y2 Mechanics Pages 148–150 Solution 35

9. A 2 kg particle is released from rest on a rough slope at angle α to horizontal, where $\tan\alpha = 2/5$. After 3 s, its speed is 6 m s^{-1}. Find the coefficient of friction μ.

Working

$\tan\alpha = \frac{2}{5} \Rightarrow \sin\alpha = \frac{2}{\sqrt{29}}, \quad \cos\alpha = \frac{5}{\sqrt{29}}.$

$u = 0, v = 6, t = 3$:

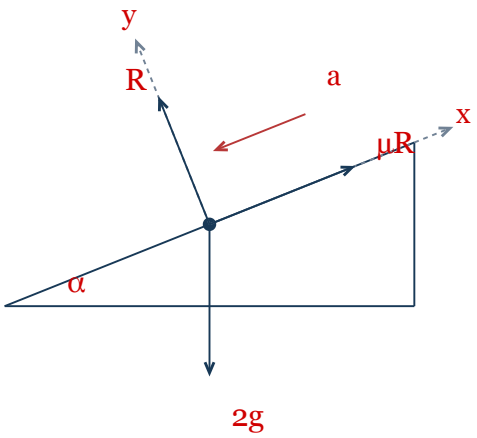
$v = u + at \quad \Rightarrow \quad 6 = 0 + 3a$

$a = \frac{6}{3} = 2\,\text{m s}^{-2}.$

Normal to plane:

$R = 2g\cos\alpha.$

Working



Working

Down the plane, $F_{\text{resultant}} = ma$:

$2g\sin\alpha - \mu R = 2 \times 2$

$2g\sin\alpha - \mu(2g\cos\alpha) = 4$

$g\sin\alpha - \mu g\cos\alpha = 2$

$\mu = \frac{g\sin\alpha - 2}{g\cos\alpha}$

$= \frac{9.8 \times 2/\sqrt{29} - 2}{9.8 \times 5/\sqrt{29}}$

$= 0.18019\ldots.$

The coefficient of friction is 0.180 (3 s.f.).

Exercise 7E — Y2 Mechanics Pages 148–150 Solution 36

11. A 5 kg particle is projected up a rough 10° slope at 16 m s^{-1}, stopping at P after 5 s. (a) Find μ. (b) State, with supporting calculations, whether it remains at rest or slides back downhill.

a

Take uphill positive.

$u = 16, v = 0, t = 5$:

$v = u + at \quad \Rightarrow \quad 0 = 16 + 5a$

$a = -\frac{16}{5}\,\text{m s}^{-2}.$

Normal to slope:

$R = 5g\cos 10°.$

Now resolve downhill, using acceleration 16/5:

$5g\sin 10° + \mu R = 5 \times \frac{16}{5}$

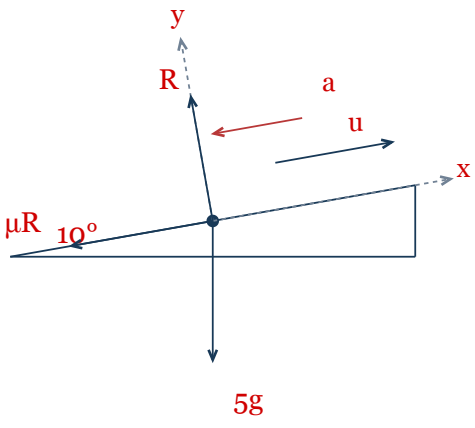
$5g\sin 10° + \mu \times 5g\cos 10° = 16$

$\mu = \frac{16 - 5g\sin 10°}{5g\cos 10°}$

$= 0.15524\ldots.$

The coefficient is 0.155 (3 s.f.).

a



b

The particle slides back if the weight component downhill exceeds the maximum available friction uphill:

$5g\sin 10° > 5\mu g\cos 10°$

$\sin 10° > \mu\cos 10°$

$0.17364\ldots > 0.15288\ldots.$

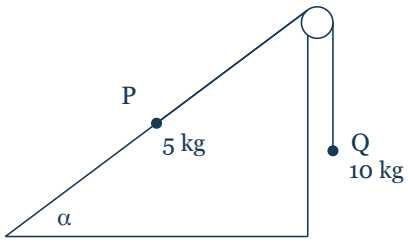
This is true, so the particle slides back downhill.

Connected Particles Involving Friction

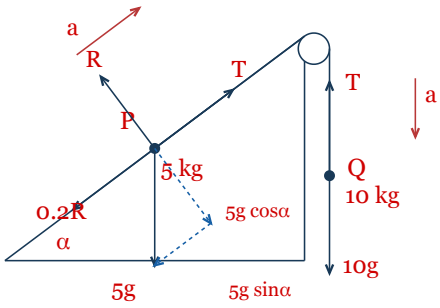
Solution 37

Connected-particle problems from Mechanics Year 1 now include friction.

Particles P, Q, masses 5 kg, 10 kg, are connected by a light inextensible string over a small smooth fixed pulley at the top of a rough plane. P rests on the plane; Q hangs freely with the string vertical and taut. The slope angle is α, with $\tan\alpha = 0.75$, and friction coefficient for P is 0.2. Released from rest, find (a) the acceleration; (b) the tension.



Diagram



a

$\sin\alpha = \frac{3}{5}, \qquad \cos\alpha = \frac{4}{5}.$

For P, normal to plane:

$R = 5g\cos\alpha = 5g \times \frac{4}{5} = 4g.$

Up the plane:

$T - 5g\sin\alpha - 0.2R = 5a$

$T - 3.8g = 5a. \qquad (1)$

For Q, downwards:

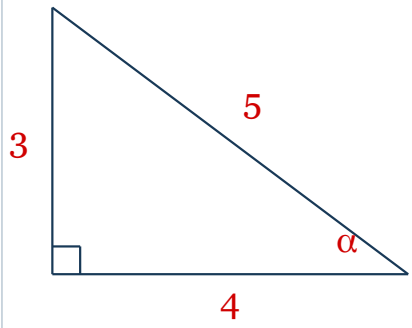
$10g - T = 10a. \qquad (2)$

Add (1),(2):

$6.2g = 15a \quad \Rightarrow \quad a = \frac{31g}{75}$

$= 4.1\,\text{m s}^{-2} \quad (2\text{ s.f.}).$

a



b

Rearrange (1):

$T = 5a + 3.8g$

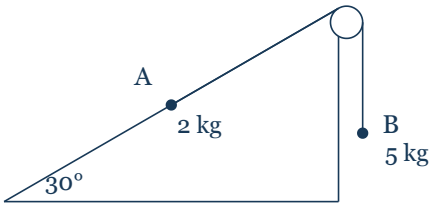
$= 57\,\text{N} \quad (2\text{ s.f.}).$

Use unrounded a.

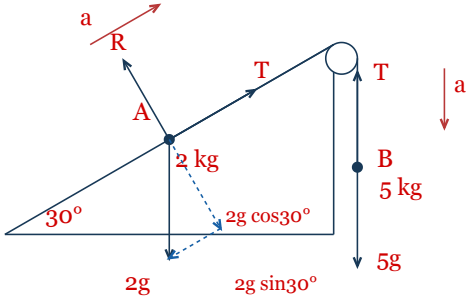
Further Example

Solution 38

A light inextensible string joins block A, mass 2 kg, on a smooth fixed 30° plane to block B, mass 5 kg, hanging freely. It lies along the greatest slope and passes over a smooth light pulley fixed at the top. Release from rest. Model the blocks as particles and ignore air resistance. (a)(i) Show B's acceleration is $4g/7$; (ii) find tension. (b) How is the inextensible-string assumption used? (c) Find the magnitude of the string's force on the pulley.



Diagram



a

For A, uphill:

$T - 2g\sin 30° = 2a$

$T = 2a + g. \qquad (1)$

For B, downwards:

$5g - T = 5a. \qquad (2)$

Combine (1),(2):

$5g = 7a + g \quad \Rightarrow \quad a = \frac{4}{7}g.$

$T = 2a + g = \frac{8}{7}g + g = \frac{15}{7}g.$

b

The taut inextensible string makes the particles' acceleration magnitudes equal.

c

Tension acts away from the pulley along each string. Choose left and down as positive; factor out constant T:

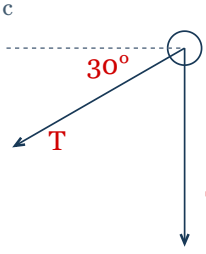
$\mathbf{F} = \begin{pmatrix} T\cos 30° \\ T + T\sin 30° \end{pmatrix}$

$= \frac{T}{2}\begin{pmatrix} \sqrt{3} \\ 3 \end{pmatrix}.$

$|\mathbf{F}| = \frac{T}{2}\sqrt{(\sqrt{3})^2 + 3^2}$

$= \frac{15}{14}g\sqrt{12} = \frac{15\sqrt{3}}{7}g\,\text{N}.$

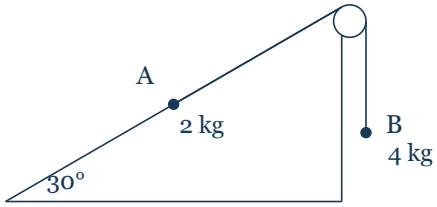
c



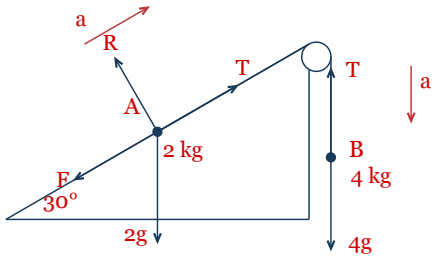
Test Your Understanding — Solution 39

Edexcel M1 (Old), May 2013(R), Q3

A rough fixed plane is inclined at 30° to horizontal. A small smooth pulley P is fixed at its top. Particles A, B, masses 2 kg, 4 kg, are joined by a light inextensible string over P. The string from A to P is parallel to the greatest slope; B hangs freely. The friction coefficient between A and the plane is $1/\sqrt{3}$. Initially held at rest, release them with string taut and A moving uphill. Find the tension immediately after release. (9 marks)



Working



Working

For B: $4g - T = 4a$.

For A: $T - F - 2g\sin 30° = 2a$.

Equivalently, for the whole system:

$4g - F - 2g\sin 30° = 6a$.

Normal to plane at A: $R = 2g\cos 30°$.

$$F = \mu R = \frac{1}{\sqrt{3}} \times 2g\cos 30° = g$$

$T - g - g = T - 2g = 2a$.

Eliminate a:

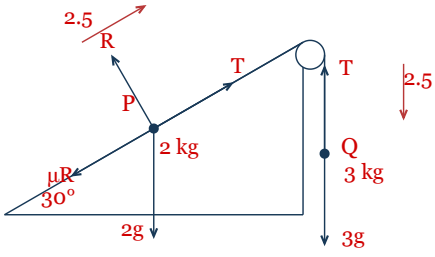
$2T - 4g = 4g - T \quad \Rightarrow \quad 3T = 8g$

$$T = \frac{8g}{3} = 26.1\,\text{N} \quad (3\text{ s.f.}).$$

Exercise 7F — Y2 Mechanics Pages 152–154 — Solution 40

3. Particles P, Q, masses 2 kg, 3 kg, are connected by a light inextensible string over a small smooth fixed pulley at the top of a rough 30° plane. P is held at rest on the plane; Q hangs freely with string vertical and taut. On release, P accelerates uphill at 2.5 m s^{-2}. Find (a) tension; (b) the coefficient of friction; (c) the string's force on the pulley.

a



a

For P, normal to plane:

$R - 2g\cos 30° = 0$

$\Rightarrow \quad R = g\sqrt{3}$.

Up the plane, $F_{\text{resultant}} = ma$:

$T - \mu R - 2g\cos 60° = 2 \times 2.5$

$T - \mu g\sqrt{3} - g = 5. \qquad (1)$

For Q, downwards:

$3g - T = 3 \times 2.5 = 7.5. \qquad (2)$

$T = 21.9\,\text{N}$.

b

Add (1),(2):

$2g - \mu g\sqrt{3} = 12.5$

$\mu g\sqrt{3} = 7.1$

$$\mu = \frac{7.1}{g\sqrt{3}} = 0.418 \quad (3\text{ s.f.}).$$

The coefficient is 0.42 (2 s.f.).

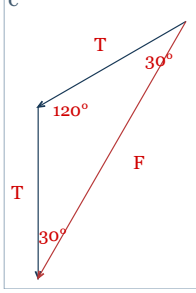
c

$F = 2T\cos 30°$

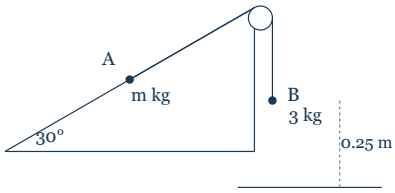
$= 43.8\cos 30° = 37.9\,\text{N} \quad (3\text{ s.f.}).$

The force magnitude is 38 N (2 s.f.).

c



4. Particles A, B, masses m kg, 3 kg, are connected by a light inextensible string over a smooth fixed pulley P. A is held on a smooth fixed 30° plane, with string AP along its greatest slope; B hangs freely. Release from rest with B 0.25 m above horizontal ground. B initially accelerates downwards at $2g/5$. A does not reach P. Find (a) tension while B descends; (b) m. B hits the ground without rebounding. (c) Find the time from impact until A reaches its highest point.



a

For B, downwards:

$$3g - T = 3 \times \frac{2}{5}g$$

$$T = 3g - \frac{6}{5}g = \frac{9}{5}g = 17.64\,\text{N}.$$

The tension is 18 N (2 s.f.).

b

For A, uphill:

$$T - mg\sin 30° = m \times \frac{2}{5}g$$

$$\frac{9}{5}g - \frac{1}{2}mg = \frac{2}{5}mg$$

$$\Rightarrow \quad m = 2.$$

c

After impact the string is slack, so tension is zero. The only force on A parallel to the plane is the downhill weight component. Taking uphill positive:

$$-mg\sin 30° = ma$$

$$\Rightarrow \quad a = -\frac{1}{2}g.$$

Now $u = 1.4, v = 0$:

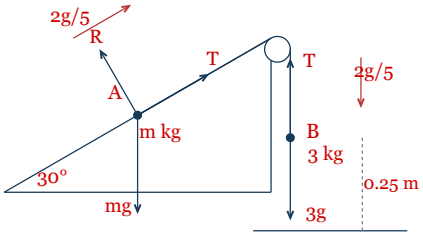
$$v = u + at$$

$$\Rightarrow \quad 0 = 1.4 - \frac{1}{2}gt$$

$$t = \frac{2.8}{9.8} = \frac{2}{7}\,\text{s}$$

$$\approx 0.29\,\text{s} \quad (2\,\text{s.f.}).$$

a



c

Before B hits the ground:

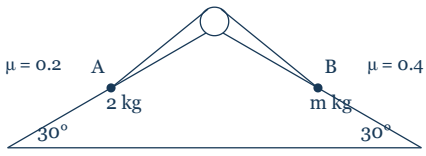
$$u = 0, \quad a = \frac{2}{5}g, \quad s = 0.25$$

$$v^2 = u^2 + 2as = \frac{4}{5}g \times 0.25$$

$$v = 1.4\,\text{m s}^{-1}.$$

5. Particles A, B on back-to-back rough 30° slopes are connected by a light inextensible string over a smooth pulley. A has mass 2 kg; B has mass m kg. The friction coefficients are 0.2 for A, 0.4 for B. Show the maximum m before motion begins is

$$\frac{10 + 2\sqrt{3}}{5 - 2\sqrt{3}}.$$



Working

Let the reactions be R_A, R_B. At maximum m, impending motion is B downhill and A uphill. For A:

$$R_A = 2g\cos 30° = \sqrt{3}g$$

$$T - 2g\sin 30° - 0.2R_A = 0$$

$$T = g + \frac{\sqrt{3}}{5}g = \left(1 + \frac{\sqrt{3}}{5}\right)g. \quad (1)$$

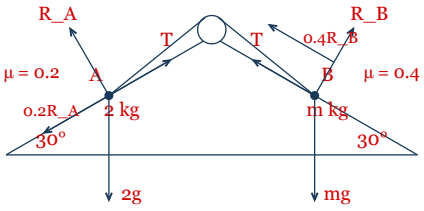
For B:

$$R_B = mg\cos 30° = \frac{\sqrt{3}}{2}mg$$

$$mg\sin 30° - T - 0.4R_B = 0$$

$$T = mg\left(\frac{1}{2} - \frac{\sqrt{3}}{5}\right). \quad (2)$$

Working



Working

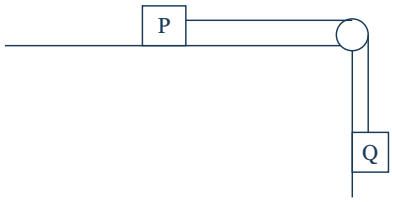
Equate (1),(2) and cancel g:

$$\left(\frac{1}{2} - \frac{\sqrt{3}}{5}\right)m = 1 + \frac{\sqrt{3}}{5}$$

$$(5 - 2\sqrt{3})m = 10 + 2\sqrt{3}$$

$$m = \frac{10 + 2\sqrt{3}}{5 - 2\sqrt{3}}.$$

6. A metal block P, mass 1.5 kg, rests on a rough horizontal workbench. A light inextensible string joins it over a small smooth pulley fixed at the edge to freely hanging box Q, mass 1.6 kg. The coefficient of friction for P is μ. Release from rest with the string taut; after 2 s, Q's speed is $6\,\text{m s}^{-1}$. Model both as particles. (a) Find acceleration of Q; (b) tension; (c) show $\mu = 0.434$ (3 s.f.); (d) how is inextensibility used?



a

$v = u + at \quad \Rightarrow \quad 6 = 0 + 2a$

$a = \dfrac{6}{2} = 3\,\text{m s}^{-2}.$

b

For Q, Newton's second law:

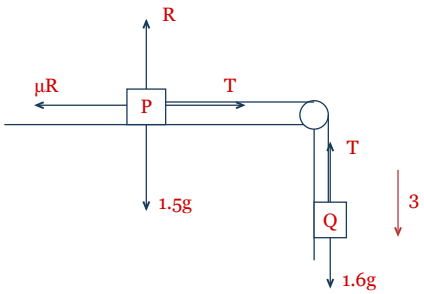
$1.6g - T = 1.6 \times 3$

$T = 1.6g - 1.6 \times 3$

$= 1.6(9.8 - 3) = 10.88\,\text{N}.$

The tension is 10.88 N.

b



c

For P, vertically:

$R = 1.5g.$

Horizontally:

$T - \mu R = ma$

$10.88 - 1.5\mu g = 1.5 \times 3$

$1.5\mu g = 10.88 - 4.5$

$\mu = \dfrac{6.38}{1.5 \times 9.8} = 0.43401\ldots$

To 3 s.f., $\mu = 0.434$.

d

The taut inextensible string gives both particles equal acceleration magnitudes. Equal tension follows from the light string and smooth pulley.